Ram Prasad Bidiyasar

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Experience

Airtel Digital — Gurugram, India
Senior Software Engineer, Xstream backend — *Jul. 2025 – Present*

- Redesigned **Continue Watching** from scratch for Xstream and IPTV: watch events keyed by user ID flow through Kafka into ScyllaDB, so concurrent and multi-device writes serialize per user; added Redis/Aerospike caching and dual-write migration, cutting sync latency by nearly 70%.
- Re-architected watch-history reads on Spring WebFlux and reactive ScyllaDB (**Watch Info v2**), cutting p99 latency by 75% and raising per-instance throughput about 4x; extended it to live TV and microdrama (REEL).
- Engineered **multi-CDN routing and failover** for IPTV playback used by 10M+ active users, evaluating MongoDB-configured rules over geo, channel, priority, weights and CDN health to cut buffering from 1% to 0.07%.
- Rebuilt the **event ingestion service** on Java 25: ∼400 to ∼2,500 req/s per pod (6x) for 80–90K events/s live-match peaks, with per-product Kafka clusters and Pub/Sub plus Kafka Connect replay so no events are lost.
- Designed and built the IPTV **session service** on multi-zone Aerospike: opaque device-bound tokens, reusable SDK validation in the SMS and playback services, and an atomic concurrent-device limit.
- Scaled page generation from 30 to 100 TPS and cut latency by nearly 30% (Java 21, gRPC, thread-pool tuning, parallel async calls); kept Home and Rush up through dependency failures with fallback retry, a kill switch and gateway CDN failover, replacing 3–4 daily spikes and 10–15 minute failures at big events (3–4 a year) with about 99% availability.
- Took over the **UCLM** campaign platform for Airtel India (20+ services, 1.5B+ messages daily): fixed build and load issues, wrote the deployment strategy, and stabilized and scaled it for the initial migration.
- Root-caused sync-service OOM kills and Scylla delete failures; resolved Black Duck and Coverity findings in 20+ services by upgrading dependencies.
- Led **device management** on Airtel Xstream: users see every device logged into their account and can log any out, so on a shared account that hits the concurrent-play limit they free a slot themselves instead of being locked out.
- Drove **partner onboarding** for 3 partners (Fridaay, Amazon Prime, Hotstar), ingesting 10,944 titles; a reusable partner-integration guide and agent rules cut integration time from 2 weeks to 3 days.
- Built an AI operations assistant (Java 25, LangChain4j, Gemini on Vertex AI) that investigates missing titles and service errors across MongoDB, ELK and Prometheus, and runs PROD fixes only after confirmation.

Airtel Digital — Gurugram, India
Software Engineer, Xstream backend — *Oct. 2023 – Jun. 2025*

- Implemented most of the code for the **AWS to GCP migration** of 20+ Xstream services: SQS/Kinesis to Kafka, S3 to GCS, MongoDB and ScyllaDB, with load-tested Kafka partitions and no downtime.
- Built the IPTV **multi-CDN rule engine** (filter, sort, rule evaluation, sync) that picks a CDN per request.
- Built a Debezium and Kafka pipeline from the Xstream catalog to data science (enrichment delay from 1–2 days to near real time) and event-driven episode promotion for watched series.
- Upgraded package and discovery services to Java 21 and Spring Boot 3.5; traced hourly 2–3 minute user-service p99 spikes to log-file rollover.
- Adapted Prebid Server (Vert.x) with a Druid analytics pipeline; built a Keycloak role-based onboarding module.

Projects

Aerospike Cluster Workspace | *Java 21, Spring Boot, React, Vite*

- Browse and edit records, check cluster health and indexes; long key-pattern operations run as cancellable jobs.

Reactive API Gateway | *Java 21, Spring Cloud Gateway, MongoDB, JWT, Docker*

- MongoDB-backed dynamic routes, JWT auth, custom filters, correlation IDs for tracing and admin APIs.

Technical Skills

Languages and frameworks: Java, C++, SQL, Spring Boot, Vert.x, Kafka Streams
Data and Messaging: Kafka, Kafka Connect, ScyllaDB, Aerospike, PostgreSQL, MongoDB, Redis, Elasticsearch, Druid, Pub/Sub
Cloud and Infra: Google Cloud Platform (GCP), Amazon Web Services (AWS), Kubernetes, Docker, Jenkins, Airflow
Practices: distributed systems, microservices, failover design, observability (Grafana, Kibana, AppDynamics)

Education

Malaviya National Institute of Technology Jaipur — Jaipur, India
B.Tech, Electronics and Communication Engineering (CGPA 8.16) — *July 2019 – May 2023*